Anderson Junior College
H2 Chemistry 9647 **2015 JC2 Prelim Paper 2 Suggested Solutions**

Section A

1 **(a)** **(i)** $AgCl$: for precipitation to occur, $[Ag^+][Cl^-] > K_{sp}\ (AgCl)$

$$\text{min } [Ag^+] = \frac{2.02 \times 10^{-10}}{0.1}$$

$$= \mathbf{2.02 \times 10^{-9}\ mol\ dm^{-3}}$$ [1]

Ag_2CrO_4: for precipitation to occur, $[Ag^+]^2\ [CrO_4^{2-}] > K_{sp}\ (Ag_2CrO_4)$

$$\text{min } [Ag^+] = \sqrt{\frac{3.01 \times 10^{-12}}{0.01}}$$

$$= \mathbf{1.73 \times 10^{-5}\ mol\ dm^{-3}}$$ [1]

(ii) beginning of titration: white ppt; end point: red ppt (accept pink) [1]

K_2CrO_4 can be used as an indicator because the $[Ag^+]$ needed to precipitate Ag_2CrO_4 is much more than that for $AgCl$ and this allows the Cl^- ions present to be sufficiently precipitated out before the Ag_2CrO_4 is precipitated out. A distinct colour change from white to red can also be observed and hence concentration of Cl^- ions can be determined accurately. [1]

(iii) (Brown) ppt of AgOH will be produced and more $AgNO_3$ will be required and this affects the accurate determination of the end point of the titration. [1]

(iv) $n(AgNO_3) = \frac{24.60}{1000} \times 0.200$

$= 0.00492$ mol

$n(Cl^-)$ in 25.0 cm^3 = 0.00492 mol

$n(Cl^-)$ in 250 cm^3 = **0.0492 mol** [1]

$n(MCl_2.nH_2O)$ in 250 cm^3 = $\frac{0.0492}{2}$

= **0.0246 mol** [1]

molar mass of $MCl_2.nH_2O$ = $\frac{6.002}{0.0246}$

= **244 g mol^{-1}** [1]

(b) **Procedure** [4]

1. Weigh a clean and dry boiling tube (or crucible).
2. Add about 5 g of **FA 1** (or any reasonable amount) into the boiling tube and weigh the boiling tube with its contents.
3. Heat the boiling tube gently for about 1 minute and then more strongly for a further four minutes.
4. Allow the boiling tube to cool before re–weighing the boiling tube and its contents.
5. Repeat the heating, cooling and weighing process until the mass of the boiling tube and its contents is consistent within 0.05 g in difference.

Results

Mass of boiling tube / g	*a*
Mass of boiling tube + **FA 1** / g	*b*
(Mass of **FA 1** used / g	$b - a$)
Mass of boiling tube + residue (after 1st heating) / g Mass of boiling tube + residue (after 2nd heating) / g	c_1 c_2
(Mass of residue / g	$c_2 - a$)

Calculation

Mass of H_2O lost = $b - c_2$
= mass of **FA 1** used – mass of residue (alternative)
= $(b - a) - (c_2 - a)$

$$n(H_2O) = \frac{b - c_2}{18.0}$$

$$n(MCl_2.nH_2O) = \frac{b - a}{244}$$

$$\frac{n(H_2O)}{n(MCl_2.nH_2O)} = \frac{n}{1}$$

$$n = \frac{244(b - c_2)}{18(b - a)}$$

$$M = 244 - 2(35.5) - 18\left[\frac{244(b - c_2)}{18(b - a)}\right]$$

$$= 173 - \frac{244(b - c_2)}{(b - a)}$$

[1]: logical sequence of steps for experimental procedure
[1]: show clear outline to obtain value of *n* and identity of *M*
[1]: heating, cooling and weighing process until consistent mass
[1]: correct table

2 **(a)** No. of potatoes = $\dfrac{300-(4\text{x}40)}{55}$ [1]

= 2.55

= **3**

(accept only 3)

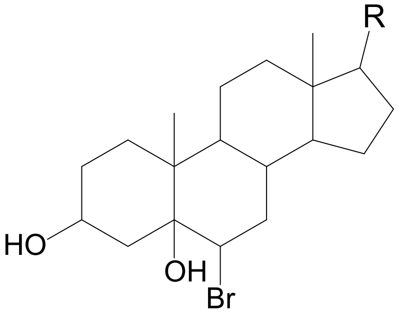
(b) **(i)** [1]

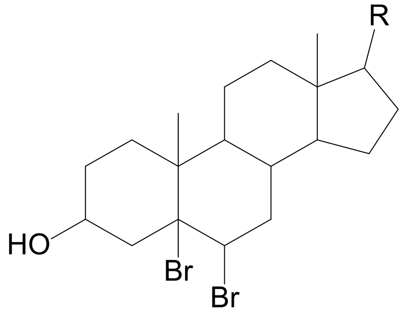
	Mg	N
mass / g	0.5 x 0.084 = 0.042	0.016
amount / mol	$\dfrac{0.042}{24.3} = 1.73 \times 10^{-3}$	$\dfrac{0.016}{(14.0)} = 1.14 \times 10^{-3}$
mole ratio	≈ 1.5	1

empirical formula of **A**: $\mathbf{Mg_3N_2}$

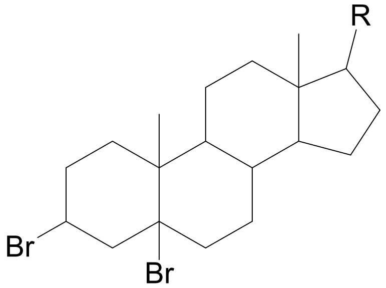
(ii) A lot of energy is needed to <u>break the strong N≡N</u> in nitrogen gas. [1]

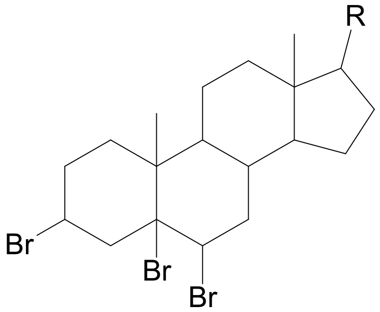
(c) **(i)** [1]



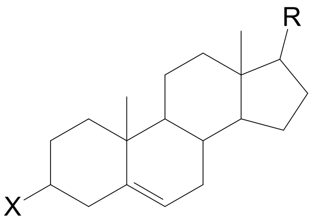
(accept )

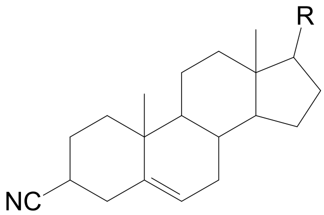
(ii) [1]

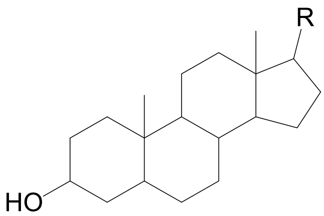


(accept  – due to presence of Br_2 from the side reaction of NaBr with conc. H_2SO_4)

(d) (i) [3]

B  (where X = C*l*, Br, I)

C 

D 

[1] each

(ii) Stage II [3]
Reagent: KCN in ethanol
Condition: heat (under reflux)

Stage IV
Reagent: $H_2(g)$
Condition: Pt, room temperature / Ni catalyst, (200 ^oC)

Stage V
Reagent: $K_2Cr_2O_7$ / $KMnO_4$, dilute H_2SO_4,
Condition: heat (under reflux)

(do not accept if stages IV and V are reversed)

(e) CaO dissolves in water to give an alkaline solution of pH 12. (any pH > 9) [3]
$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq)$

$CaCl_2$ dissolves in water to give a neutral solution pH 7.
$CaCl_2(s) + aq \longrightarrow CaCl_2(aq)$

(accept dissolves with slight hydrolysis to give a weakly acidic solution of pH 6.6)
(6.5 < any pH < 7)
$CaCl_2(s) + aq \longrightarrow CaCl_2(aq)$
$Ca(H_2O)_6^{2+} + H_2O \rightleftharpoons Ca(OH)(H_2O)_5^{+} + H_3O^{+}$

[1]: correct description for both with the corresponding equation, if any
[1]: each correct pH (must give specific value)

3 **(a)** **(i)** $pCO_2(g) = 101.3 \times (3.5 \times 10^{-4})$ [2]

$= \mathbf{0.0355\ kPa}$

$$K = \frac{[CO_2(aq)]}{pCO_2(g)}$$

$[CO_2(aq)] = 3.3 \times 10^{-4} \times 0.0355$

$= \mathbf{1.17 \times 10^{-5}\ mol\ dm^{-3}}$

[1]: correct partial pressure of CO_2
[1]: correct calculated value of solubility (BOD if non-sequential)

(ii) $$K_c = \frac{[H_2CO_3]}{[CO_2]}$$ [1]

$[H_2CO_3] = 1.7 \times 10^{-3} \times 1.17 \times 10^{-5}$

$= \mathbf{1.99 \times 10^{-8}\ mol\ dm^{-3}}$

(b) $$K_{a1} = \frac{[H^+]\,[HCO_3^-]}{[H_2CO_3]}$$ [2]

$[H^+]$ dissociated by H_2CO_3

$= \sqrt{(4.5\text{x}10^{-7})(1.99\text{x}10^{-8})}$

$= \mathbf{9.46 \times 10^{-8}\ mol\ dm^{-3}}$

$[H^+]_{total} = 9.46 \times 10^{-8} + 10^{-7}$ (from auto-ionisation of water)

$= 1.95 \times 10^{-7}$

$pH = -\log_{10} 1.95 \times 10^{-7}$

$= \mathbf{6.71}$

[1]: correct $[H^+]$ dissociated by H_2CO_3
[1]: correct pH

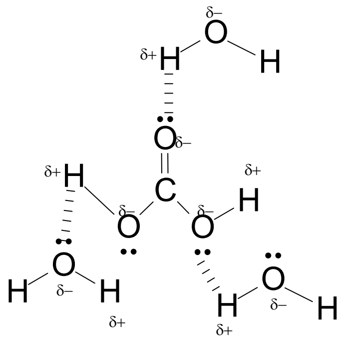
(c) When organic matter decay, carbon dioxide gas is released, $[CO_2(g)]$ increases. [1]

Position of equilibrium (P.O.E.) for reaction 1 shifts right to decrease the $[CO_2(g)]$, resulting in an increase in $[CO_2(aq)]$. Thus, P.O.E. for reaction 2 shifts right to decrease $[CO_2(aq)]$. This results in an increase in $[H_2CO_3]$ which in turn causes the P.O.E. for reaction 3 to shift right. [1]

This result in an increase in $[H^+]$. Thus, pH decreases. [1]

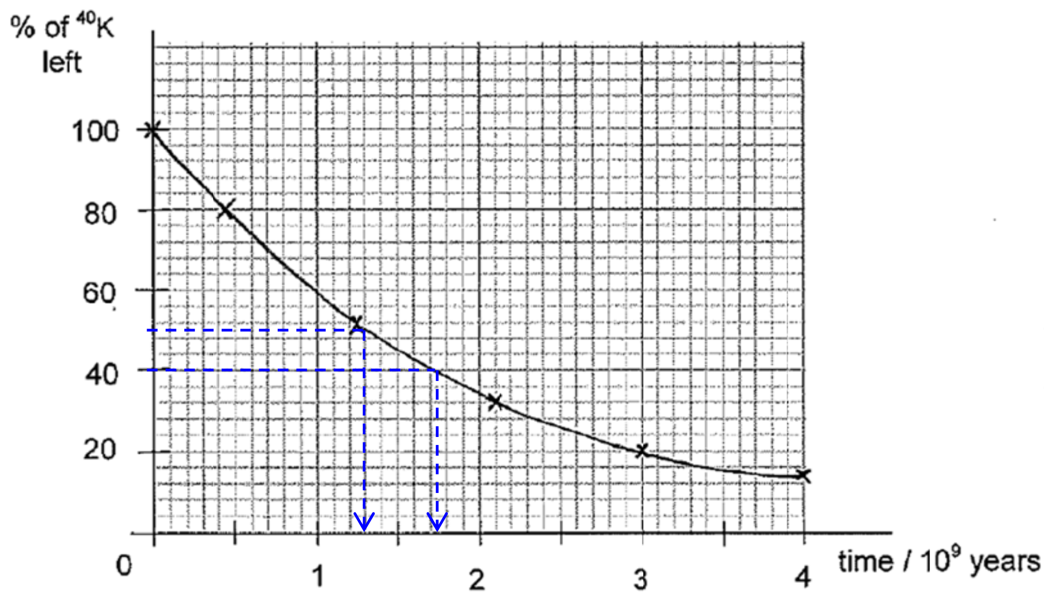
(d) **(i)** (intermolecular) hydrogen bonding [1]

(ii) [3]



[1]: structure of carbonic acid
[1]: 3 water molecules, 1 around C=O oxygen, 1 around each –O–H group
(accept HB to either O or H of H_2CO_3)
[1]: correct drawing of hydrogen bonding (lone pairs, dipoles shown)

4 **(a)** **(i)** [1]



[1]: correctly plotted points (check 2nd, 3rd and 4th points) with a smooth curve

(ii) From the graph, [2]

time taken when 100% of ^{40}K is reduced to 50% = **1.3×10^9 year**

time taken when 80% of ^{40}K is reduced to 40% = $(1.75 - 0.45) \times 10^9$

= **1.3×10^9 year**

$t_{½}$ is (relatively) constant, therefore the order of reaction of this decay is first order.

[1]: correct determination of half−life (shown on grid / in working)

[1]: $t_{½}$ is constant ($\pm 0.1\times10^9$ year)

Comments:

Although it seems possible to find 2 consecutive half-life from the graph (100% $\longrightarrow$ 50% $\longrightarrow$ 25%), it is not advisable because you are not likely to read off 25% accurately from the graph due to the scale used.

(b) **(i)** $n_o = (128.5 + 77.1) \times 10^{18}$ [1]

$= \mathbf{205.6 \times 10^{18}}$ **atoms**

(ii) When ^{40}K content reaches 77.1×10^{18}, it has undergone n half–life. [1]

$$1 \longrightarrow \frac{1}{2} \longrightarrow \frac{1}{4} \longrightarrow \frac{1}{8} \longrightarrow \quad \longrightarrow [(\frac{1}{2})^n]$$

Hence $\dfrac{77.1\text{x}10^{18}}{205.6\text{x}10^{18}} = (\frac{1}{2})^n$, n = no. of half–life

$n = \lg(\dfrac{77.1}{205.6}) \div \lg(\dfrac{1}{2})$ (accept ln)

≈ 1.42

Time taken for mica to be left with 77.1×10^{18} atoms of ^{40}K

$= 1.42 \times 1.3 \times 10^9$

= **1.84×10^9 year**

(accept reading from graph where % of ^{40}K left = $\dfrac{77.1x10^{18}}{205.6x10^{18}}$ x 100% = 37.5%)

Alternative

1 $\longrightarrow$ 0.5 $\longrightarrow$ 0.25 (<0.375)

Time taken for mica to be left with 77.1 x 10^{18} atoms of ^{40}K

≈ 2 x 1.3 x 10^9 / ≈ 1.5 x 1.3 x 10^9

= 2.6 x 10^9 year / 1.95 x 10^9 year

(c) Neutrons are uncharged and are not repelled by protons in the nuclei. [1]

(d) (i) [1]

Particle	Mass number	Number of		
		protons	electrons	neutrons
L^{2+}	111	**49**	47	**62**
M^{2-}	51	**25**	27	**26**

(ii) Must be less than sum of 1st IE + 2nd IE of Ga = 577 + 1980 = 2557 kJ mol^{-1} [1]

[1]: any specific value < 2557 (must be shown)

In is below Ga in the Period Table so the sum of its 1st and 2nd ionisation energies is less than that of Ga. As the no. of quantum shell increases down the group / valence electrons of In are further away from the nucleus and experience less electrostatic attraction by the nucleus, hence requiring less energy to remove it. [1]

(accept explanation in terms of across a Period using data for Sr or Sn)

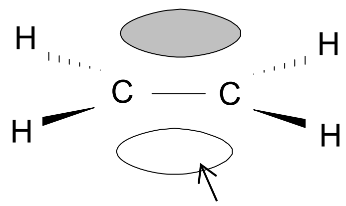
(iii) angle of deflection $\alpha \ \dfrac{\text{charge}}{\text{mass}} (\dfrac{z}{m})$ [2]

Particle	z/m ratio	angle of deflection
Ca^{2+}	$\dfrac{+2}{40}$	+10.0°
M^{2-}	$\dfrac{-2}{51}$	$[\dfrac{-2}{51} \div \dfrac{+2}{40}]$ x (+10.0°) = **–7.84°**

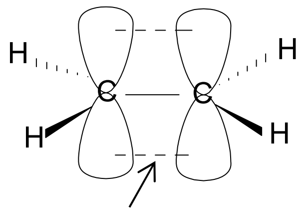
[1]: correct sign (u/c)

[1]: correct magnitude with working

5 (a) (i) [1]



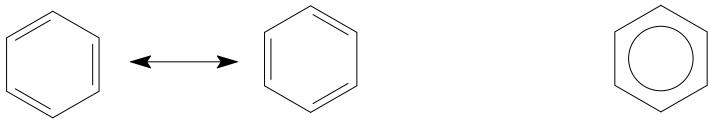
OR



(two 2p orbitals overlapping side–on to form π bond)

(ii) The π bond is described as a localised π bond because the <u>electron density of the π bond is shared between 2 adjacent C atoms only</u>. [1]

(iii) Benzene exists in these 2 resonance structures and hence to show the delocalisation of the π electron cloud, a resonance hybrid can be represented as shown below. [1]



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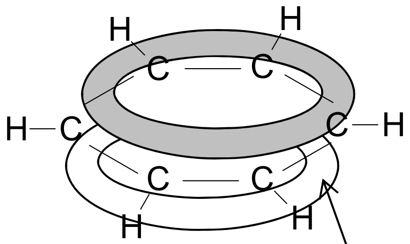

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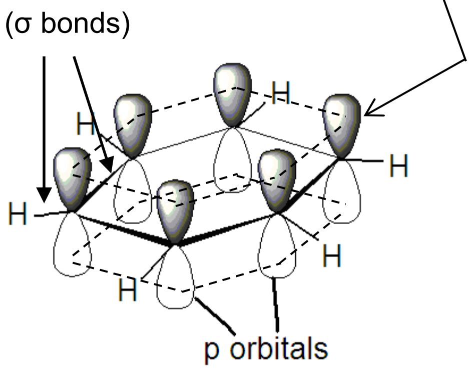


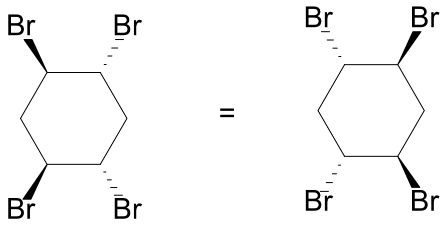
<u>continuous overlap of p–orbitals</u> results in a <u>delocalised π electron cloud above and below</u> the plane of molecule.

Delocalised π bond in benzene refers to the <u>equal sharing of the 6 π electron among all the 6 C atoms in benzene</u> instead of having each of the 3 pairs of the π electron between any 2 C atoms. [1]

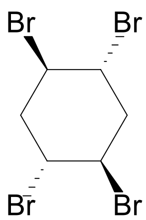
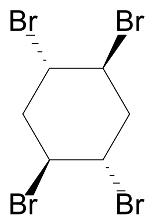
OR

Delocalised π bond is present in benzene because <u>each pair of the π electrons is not localised but shared between more than 2 adjacent C atoms</u>.

(iv) The actual value for the enthalpy change of hydrogenation of benzene is less than expected as the structure of benzene is resonance <u>stabilised</u> due to <u>delocalisation</u> of the <u>π electrons</u>. [1]
Hence, <u>more energy than expected</u> is needed to <u>break the (pi) bond(s)</u> in benzene when it reacts with H_2. [1]

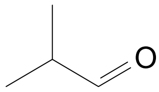
(b)  [3]

(due to presence of plane of symmetry)

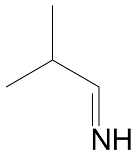
 

[1] each stereoisomer
first mark only awarded if just one of the two structures is given (since they are the same) or if two given structures are identified as identical.

6 **(a)** **(i)** [2]



Q

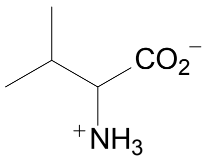


R

Not required
Q does not react with sodium.
⇒ absence of alcohol or presence of aldehyde/ketone
R does not decolourise $Br_2(aq)$.
⇒ absence of C=C
⇒ likely removal of H and OH from adjacent N and C atoms

(ii) nucleophilic addition [1]

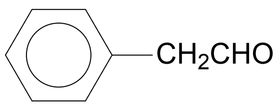
(iii) [1]



S exists in zwitterionic / ionic structure.

[1]: correct structure and drawing

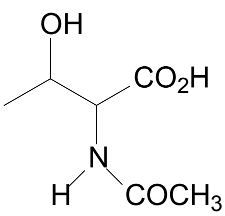
(iv) [1]



(b) [3]

'no reaction'

T



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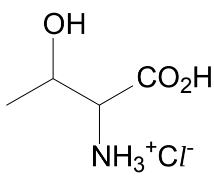

V

[1] each

(c) van der Waals' forces lost / broken / disrupted when phenylalanine is substituted with threonine which is able to form hydrogen bonding [1] [1]

(d) **(i)** **3** [1]

(ii) **hydrolysis** [1]

(iii) [1]